MAKERERE UNIVERSITY

COLLEGE OF COMPUTING AND INFORMATION SCIENCES

DEPARTMENT OF COMPUTER SCIENCE / SOFTWARE ENGINEERING

BSE4104: Emerging Trends in Software Engineering

WEEK 8 PROGRESS REPORT

(FINAL ENGINEERING CLOSEOUT)

Project: University Student-Support Case Agent

Group: BSE4104 Capstone Group — Student Support AI Agent

1. Project Details

NAMES	REGISTRATION NO:	ROLE
KABANDA ARTHUR	23/U/08510/PS	Project & Requirements Lead
TUMUKUNDE KATO ANDREW	23/U/18082/PS	Application & Integration Lead
ARIKO SOSSY JOEL	23/U/06399/EVE	AI Engineering Lead
GARANGA JOHN	23/X/22975/PS	Quality, Security & DevOps Lead

2. Work Completed Against Weekly Objectives

The primary objective for Week 8 was final system hardening, complete multi-layer regression testing, closure of the Week 7 failure catalogue, and project verification sign-off.

- **Full-Suite Automated Verification:** Executed the complete test suite spanning all 9 architectural layers across 18 test modules. Out of 502 total items, **496 passed**, **6 were safely skipped** (live Gemini quota-dependent tests), and **0 regressions** were detected in 19.53 seconds.
- **Closure of Failure Catalogue (G1–G6):** Formally validated and closed all 6 high-priority vulnerabilities identified in Week 7 with deterministic regression tests in tests/test_guardrails.py and tests/test_stop_conditions.py.
- **Human-in-the-Loop & Boundary Enforcement:** Verified that examination-related disputes and high-urgency requests remain gated as PENDING_STAFF_APPROVAL, and malicious requests to alter grades or financial records are blocked deterministically.
- **MCP Protocol Conformance:** Validated the JSON-RPC 2.0 stdio protocol adapter (src/mcp_adapter.py) ensuring external MCP client compatibility and error parity.

3. Key Engineering Decisions and Rationale

1. **Deterministic Application-Layer Guardrails over Prompt-Only Controls**

   Decision: Implemented strict ASCII regex validation, topic blocking (DISALLOWED_TOPIC), and sensitive credential redaction (_redact_secrets()) inside src/tools/handlers.py rather than relying on LLM system prompt instructions.

   Rationale: LLM prompt following is non-deterministic and susceptible to jailbreaking or phrasing evasion; deterministic code boundaries provide 100% enforceable safety guarantees.

2. **Hard Strikeout Bounds in Orchestrator Loop**

   Decision: Configured the orchestrator (src/orchestrator.py) to halt after 2 consecutive tool/retrieval failures with an explicit TOOL_ERROR stop state.

   Rationale: Prevents infinite agent replanning loops, token budget exhaustion, and hallucinated answers during cascading tool failures.

3. **Exponential Backoff & Origin-Preserving Outage Reporting**

   Decision: Added automated retry backoff in src/model_client.py for transient 429/503 errors and returning a structured HTTP 502 Bad Gateway payload with complete diagnostic attempt histories during persistent outages.

   Rationale: Differentiates transient network/rate spikes from genuine software bugs while providing clear observability traces.

4. **Synthetic Scoping for Pre-Production Safety**

   Decision: Maintained synthetic mock stores and cookie sessions, explicitly declaring that student identity authentication is an upstream requirement prior to real-world deployment.

   Rationale: Protects real student records from unintended exposure during capstone testing and academic evaluation.

4. Failures, Challenges, and Current Response

Challenge / Observed Failure	Root Cause	Engineering Response & Retest Status
Adversarial Outcome Rewording (G1)	Evasion of grade/fee alteration blocks via synonyms (fix, scrub, zero).	Expanded regex verb/protected-subject filters in src/tools/handlers.py. Retested: 100% of 18 adversarial variants return HTTP 403.
Volunteered Credentials in Free Text (G2)	Passwords and PINs included in ticket text were written directly to storage.	Added _redact_secrets() at tool entry boundary to mask credentials as [REDACTED_SECRET]. Verified in test_g2_credentials_are_never_persisted.
Unicode / Non-ASCII Digits (G3)	Full-width numerals (e.g. CAS-２０２６-００１) passed \d regex and failed as CASE_NOT_FOUND.	Enforced strict ASCII [0-9] regular expressions in CASE_ID_RE. Retested: All non-ASCII inputs return HTTP 400 INVALID_ID_FORMAT.
Live Model Quota Limits (Gemini 429)	Rate limit exhaustion during batch automated testing of live endpoints.	Gated live model tests under @pytest.mark.skipif, allowing full 496-case regression harness to execute deterministically offline in 19.53s.

5. Repository and ClickUp Evidence Links

- **Main Git Repository:** https://github.com/Kabandaarthur/BSE4104
- **ClickUp Sprint Board:** https://app.clickup.com/90152696819/v/o/s/901512235988

- **Key PRs & Commits:**
  - PR #27 (a731b3a / 82ed081): Week 7 evaluation, guardrail tests, and failure catalogue.
  - PR #26 (1318d27 / 92a49f6): Structured timing and trace capture across pipeline.
  - PR #25 (4a84ff2): Boundary hardening and adversarial test suites.
  - Commit 689ee04: Web UI chat integration and state binding.
  - Test Findings Document: docs/evaluation/WEEK_8_TEST_FINDINGS.md

6. Individual Contribution Summary

Team Member & Role	Tasks Owned	Verifiable Evidence
Kabanda Arthur *(Project & Requirements Lead)*	Overall sprint coordination, requirement traceability matrix, HITL authorization policy enforcement, ClickUp task auditing, and final report synthesis.	• README.md updates • Weekly Reports. • test/.... • Review all PR made • src/static/index.html (web ui)
Tumukunde Kato Andrew *(Application & Integration Lead)*	Session store hardening, atomic file-replacement persistence, 24-hr session TTL eviction, FastAPI web endpoints, and MCP JSON-RPC stdio protocol adapter.	• src/session_store.py • src/mcp_adapter.py • tests/test_session_store.py • tests/test_mcp_adapter.py
Ariko Sossy Joel *(AI Engineering Lead)*	Foundation model client resilience, exponential retry backoff, RAG vector query bounds, and source citation provenance formatting in Chroma store.	• src/model_client.py • src/retrieval.py • tests/test_model_client.py • tests/test_retrieval.py
Garanga John *(Quality, Security & DevOps Lead)*	Executed full 502-test regression harness, closed G1–G6 failure catalogue retests, implemented _redact_secrets(), ASCII regex enforcement, and test findings documentation.	• src/tools/handlers.py • tests/test_guardrails.py • tests/test_tools.py • docs/evaluation/WEEK_8_TEST_FINDINGS.md